

Machine Learning-Based Intrusion Detection System for Network Traffic Classification

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Abstract—Protecting networks and systems from cyber attacks and unauthorized access is important. This Project is demonstrate Machine Learning Based Intrusion Detection System. It focus on identify suspicious activities and potential security attacks within a network. Developed IDS used machine learning algorithms such as Random Forest and XGBoost. They analyze features of network traffic and decide whether normal or anomaly behavior. For data preprocessing and model development Pandas and Scikit-Learn Libraries were used. The system performance was evaluated using accuracy, Precision value, recall value and false positive rate. This project demonstrates the value of python and machine learning in cyber security with highlighting the role of IDS in improving network security and detecting cyber threats. (*Abstract*)

Keywords—IDS, Random Forest, XGBoost, Scikit-Learn, pandas (*key words*)

I. INTRODUCTION

At the present cyber threats like malware, unauthorized access, denial of services attack and other malicious activities are increased. For detecting modern digital attacks are difficult by traditional security mechanisms like firewalls. Because of that IDS is very important in monitoring network traffic and identifying suspicious behavior in real time. With the development of Artificial Intelligence and Machine learning technologies IDS solutions have become more effective in detecting anomalies and improving cyber defenses.

This System developed as Machine Learning Based Intrusion Detection System using python and Scikit-learn. It is designed to detect network traffic with classifying as normal or anomaly. Used algorithms are Random Forest and XGBoost. 41 features and a class label which indicates as normal or anomaly are contained in the dataset which is downloaded from Kaggle. Among the 42 features, 13 important features were selected.

Two machine models were developed. Model 1 developed using Random forest classifier and Model 2 developed using XGBoost. For model reliability, cross validation, classification reports and confusion matrices were used. Developed web based interface integrating with two models allows users to enter their inputs for detecting the behavior of the traffic. It also integrate with AbuseIPDB to check the reputation of source IP addresses and alert users about potentially malicious IPs.

II. LITERATURE REVIEW

Machine learning for the cyber security is mostly study area at the present. Because of that How to develop IDS using machine learning also study by researchers. This review focus on the use of Random Forest, XGBoost and common issues like class imbalance and feature selection.

To reduce the variance and bias of individual classifiers, ensemble methods are used. In a comprehensive comparative study, Bibers et al. [1] evaluated 14 different individual and ensemble models on the CICIDS-2017 and RoEduNet SIMARGL2021 datasets. Researchers found that ensemble methods, Random forest achieved perfect scores, significantly outperforming individual models like Logistic Regression and Multi Layer Perceptron. This finding aligns with this system , where RF is a primary classifier.

In the IDS Random Forest is widely used because of the robustness and ability to handle high dimensional data. Waskle et al. [5] used Principal Component Analysis for feature reduction with a Random Forest classifier. It demonstrating improved detection rates on benchmark datasets. As well as Balyan et al. [4] proposed a hybrid model using an improved random forest method with evolutionary optimization, achieving high accuracy.

XGBoost, known for its speed and performance, it has also gained traction. Dhaliwali et al. [7] developed XGBoost for intrusion detection. It also highlight its efficiency and accuracy in handling missing values and feature importance. Recently Koyyagura et al. [8] studied deep learning architectures (CNN, LSTM, GRU) for IDS achieving up to 98.59% accuracy.

The class imbalance problem is major issue in IDS, where attack samples are often rare. Kurniabudi et al. [6] performed an extensive feature analysis of the CICIDS-2017 dataset, showing that without proper balancing, models become biased towards the majority (normal) class. Feature selection also a critical issue. Das et al. [9] focused that combining ensemble machine learning with feature selection significantly improves performance and reduces computational overhead. Dropped features based on the correlation and select 13 features of 42 features is strongly supported by these findings.

In some studies researchers address the practical deployment and usability of ML-based IDS. Most research concludes with performance metrics. Alotaibi and Ilyas [10] focus ensemble learning framework for IoT devices and web based tools mentioned my Kim et al.[11] focus of how to create usable security tools. In the system's web interface integrated with the AbuseIPDB API for real time IP reputation checks directly

addresses this gap, moving from theoretical performance to practical application. Literature review demonstrates the efficiency of RF and XGBoost for IDS, addressing class imbalance and feature redundancy and the value of deploying models user based tools.

III. METHODOLOGY

Methodology of this research contain of data collection, preprocessing, feature engineering, model training, evaluation and deployment.

A. Dataset Collection

Data set is found from Kaggle. It is already divided in to training and testing datasets. It is contained 42 features according to a class column representing normal and anomaly network traffic behavior. It has Normal class 13449 of records and Anomaly class 11743 of records. The data set was imbalanced. Because of that balancing techniques were applied to ensure equal quantity of both classes. So balanced dataset includes 11743 Normal records and Anomaly records.

B. Data Preprocessing

Using libraries like Pandas, Numpy and Scikit learn data preprocessing part was performed. Before training the model, checked missing values, Duplicate records and correlation between features. No null or duplicated data were found in the data set. In the correlation values two columns were NaN. Because of that those two columns were dropped from the data set which were num_outbound_cmds and is_host_login. To convert the categorical class labels in to numerical values, (Anomaly = 0, Normal =1) Label Encoding was used.

C. Feature Selection

By analysing correlations of 42 features, select only 13 features based on the correlation values. Important features were selected because they are highly affect to the result anomaly or normal behaviour. Because of the selecting only 13 features, reduced the computational complexity.

D. Data Visualization

Before training the model, dataset was visualized to analyze class distribution, feature correlation, data behavior patterns. Visualization techniques helped to identify relationships between features and detect imbalance issues within the dataset.

E. Model Training

Two Models were developed using Scikit-learn, Python and Google colab.

- Model 1 : Random Forest Classifier

The first model was trained using the Random Forest algorithm due to its robustness and high accuracy in classification tasks.

- Model 2: XGBoost

The second model was trained using the XGBoost algorithm to compare performance with the Random Forest model.

F. Model Evaluation

The model were evaluated using Accuracy score, Cross validation, Classification report and Confusion matrix. The performance results: Random Forest had 99.39% accuracy and XGBoost had 98.40% accuracy. To reduce

overfitting tasks and verify model reliability, cross validation was performed.

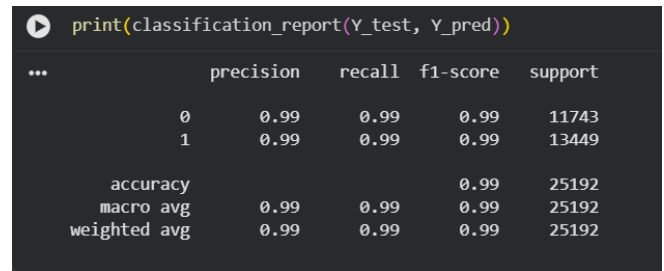
G. Web-Based Interface Development

Web based IDS interface was developed because of user to enter selected features manually, select the predict model, Identify Abuse IP addresses and perform real time intrusion detection.

IV. RESULTS AND DISCUSSION

The results shows that machine learning techniques are highly effective for intrusion detection tasks. Random Forest Classifier achieved the highest performance with an accuracy of 99.39% XGBoost model achieved 98.40% accuracy.

The Random Forest model produced highly balanced results for both anomaly and normal traffic classes. The classification report showed precision, recall and F1-score values of 0.99 for



```
print(classification_report(y_test, y_pred))
```

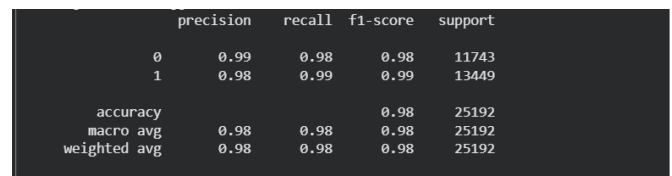
	precision	recall	f1-score	support
0	0.99	0.99	0.99	11743
1	0.99	0.99	0.99	13449
accuracy			0.99	25192
macro avg	0.99	0.99	0.99	25192
weighted avg	0.99	0.99	0.99	25192

both classes, which indicates excellent detection capability and minimal misclassification.

Fig. 1. Random Forest Classification Report

Model has high recall value and it confirms that the Random Forest model successfully identified the most of the anomalous traffics without missing any attack patterns.

The XGBoost model also showed strong performance with high precision and recall values. But its performance was lower than random forest model, it still achieved reliable intrusion detection results.



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Fig. 2. XGBoost Classification Report

The results demonstrates that data balancing, feature selection, scaling and data preprocessing improved model performance. It had 42 features. But training the model with all of these features is more complexity because need more computational power. Because of the from the finding correlation values features were reduced to 13.

Cross validation and confusion matrix analysis confirmed the consistency and reliability of both models. The developed web based interface also enhanced the practical implementation of IDS by allowing users to manually input selected feature values and choose preferred prediction model. Furthermore,

the integration of AbuseIPDB enabled source IP reputation analysis, providing additional security awareness by malicious IP detection and alert generation.

V. CONCLUSION

To identify anomalous network traffic Machine Learning Based Intrusion Detection System is the best and this research successfully developed it. This study demonstrates how the machine learning techniques can be used for identifying cyber threats with high accuracy. Data preprocessing, feature selection, balancing techniques and scaling improved model performance. For increase the training efficiency and detection capability, 13 important features were selected from 42 features. Random Forest Classifier had 99.39% accuracy and XGBoost model had 98.40% accuracy. For increasing effectiveness and reliability of both models, performed cross-validation, classification report and confusion matrices. By integrating AbuseIPDB to web interface performed reputation analysis were performed due to enhanced the practical implementation of the system by enabling real time intrusion detection and malicious IP verification. Overall, this research highlights the potential of python, scikit learn and machine learning technologies in developing intelligent cyber security solutions capable of protecting networks against evolving cyber threats

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